

Moves on the Space of Bilinear Schemes: a Field Guide to Rank-Hunting Search

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Reproducible methods note — logic repo, matmul track, 2026-07-14. Companion to the flower, adds, 53-schemes, and zkML papers; all engines cited live in this repository.

1 The space

The general **format** is $\langle m, n, k \rangle$: the tensor of the bilinear map that multiplies an $m \times n$ matrix by an $n \times k$ matrix (rank-one terms then have $a_t \in K^{m \times n}$, $b_t \in K^{n \times k}$, $c_t \in K^{m \times k}$; the literature also writes (n, m, p)). This note works throughout with the square diagonal $\langle m, m, m \rangle$ — the case that recurses on itself — so fix $\langle m, m, m \rangle$ and a coefficient field K . A **scheme of rank** r is a decomposition of the matrix-multiplication tensor into r rank-one terms,

$$T_m = \sum_{t=1}^r a_t \otimes b_t \otimes c_t, \quad a_t, b_t, c_t \in K^{m \times m},$$

verified by the m^6 Brent equations. The **space** is the union of rank strata: at 3×3 the stratum $r = 27$ contains the naive scheme, $r = 23$ the records (17,376+ known classes over $\mathbb{Z}$), $r = 22$ nothing known; at 4×4 the strata run 64 (naive), 49 (Strassen²), 48 (dyadic records over odd characteristic), 47 (characteristic 2 only). Two schemes are **equivalent** (de Groote) if a sandwich (PaQ^{-1} , QbR^{-1} , $R\bar{c}P^{-1}$), a cyclic slot rotation, or a per-summand rescaling maps one to the other; searches operate on gauge-canonical representatives (monic slots over $\mathbb{F}_p$; primitive sign-normalized over $\mathbb{Q}$).

What is known about the *shape* of the space, from the literature and from this repository’s certificates:

- **It is disconnected at bounded rank.** Kauers–Moosbauer’s walks from the 3×3 naive scheme land in 584 distinct connected components of the rank- ≤ 23 flip graph (64,061 vertices) [1]. Our 4×4 fringe certificate is a **component-rigidity** certificate: the rank-48 seed’s component under solved dyadic moves at rank ≤ 50 is *finite* — 920 states in 16 clusters hanging off the seed as a cut vertex, exhaustively enumerated and closed.
- **Altitude connects, but slowly.** Splits (rank+1 excursions) merge components: our chase explored eight levels above the 4×4 seed (1.45B states) without meeting a second rank-48 class.
- **The field changes the geometry.** Over $\mathbb{F}_2$ the factor value space has 2^{m^2} points, so equal factors and coplanar triples arise by birthday collision as a walk churns; over a 64-bit proof field the value space is effectively infinite, collisions never occur, and only *algebraically solved* moves create the coincidences that descents spend (measured: 448M flips with random and small-dyadic λ from naive over Goldilocks produced zero reductions; the same protocol over $\mathbb{F}_2$ produced ~ 60 within minutes).

Terminology used throughout the program, in two strengths: **radius rigidity** — no smaller-rank scheme exists within replacement distance k of a seed (bought by exhaustive repair ladders, M-8); and **component rigidity** — the seed’s *entire* move-component is finite, enumerated, and contains no smaller rank (bought by closure/BFS certificates; strictly stronger, and it delivers the component’s anatomy as a bonus). “Rigid” without qualification in our companion papers means one of these two, and the certificate type is always stated.

2 The moves

Notation for the toy examples: e_{ij} is the matrix unit; the naive 2×2 scheme has eight summands $e_{ik} \otimes e_{kj} \otimes e_{ij}$; Strassen's seven summands are labeled $M_1..M_7$ ($M_1 = (e_{11}+e_{22}) \otimes (e_{11}+e_{22}) \otimes (e_{11}+e_{22})$, $M_2 = (e_{21}+e_{22}) \otimes e_{11} \otimes (e_{21}-e_{22})$, $M_6 = (e_{21}-e_{11}) \otimes (e_{11}+e_{12}) \otimes e_{22}$, ...).

M-1. Flip (rank-preserving; the K–M edge). Two summands whose slot- s factors are *equal* (post-gauge) exchange material in the other slots: with shared a and transfer λ ,

$$u \otimes b_1 \otimes c_1 + u \otimes b_2 \otimes c_2 \rightarrow u \otimes (b_1 + \lambda b_2) \otimes c_1 + u \otimes b_2 \otimes (c_2 - \lambda c_1).$$

Toy (naive 2×2 , summands $(1, 1, 1)$ and $(1, 1, 2)$, both with $a = e_{11}$, $\lambda = 1$): $b_1 \leftarrow e_{11} + e_{12}$ and $c_2 \leftarrow e_{12} - e_{11}$; the two cross terms cancel and the sum is unchanged. Over $\mathbb{F}_2$, $\lambda = 1$ is the only choice; over $\mathbb{Q}$ we restrict λ to dyadics (a genuine legality constraint: arbitrary rationals explode denominators through repeated flips); over $\mathbb{F}_p$ any $\lambda \neq 0$ is legal — and “non-integer” has no meaning there, since $\mathbb{Z}$ surjects onto $\mathbb{F}_p$: the element $\frac{1}{2} = (p+1)/2$ is as much an integer residue as 2, and residues that are images of no small rational are equally available. The distinctions that matter over $\mathbb{F}_p$ are strategic, not legal: small-image λ preserve coincidence structure, solved λ create it, generic λ destroy it (§1).

M-2. Reduction (rank -1 , and the characteristic-2 -2). Two summands sharing *two* slots merge: $u \otimes v \otimes c_1 + u \otimes v \otimes c_2 = u \otimes v \otimes (c_1 + c_2)$; if $c_1 + c_2 = 0$ both vanish — over $\mathbb{F}_2$ this fires whenever two summands become fully identical, a rank -2 event with no analogue over large fields. The $\mathbb{Q}/\mathbb{F}_p$ variant also merges shared-slot pairs whose third slots are *proportional*. Reductions are the only rank-lowering moves.

M-3. Split (rank $+1$). Rewrite $f_i = \mu f_k + (f_i - \mu f_k)$ in one slot, producing a summand that *shares* that slot with summand k . Toy ($\mu=1$, slot a , M_1 against M_2): $a(M_1) = e_{11} + e_{22} \rightarrow$ parts $(e_{21}+e_{22})$ and $(e_{11}-e_{21})$, each tensored with $b(M_1) \otimes c(M_1)$. Splits buy altitude and manufacture shared pairs (the flip fuel). A fresh split's two parts share *both* other slots, so a reduce-on-sight walk must be gated (the split/merge livelock found building pursue9: reductions legal only after an intervening flip).

M-4. Solved flip (coincidence-targeted; ours). For a shared pair (i, j) and a third summand m , solve $f_i(t) + \lambda f_j(t) = \mu f_m(t)$ by 2×2 Cramer in the transfer slot t : the flip with *that* λ makes summand i 's t -factor proportional to m 's — an alignment created by algebra, not luck. Toy: $T_1 = u \otimes b_1 \otimes c_1$, $T_2 = u \otimes b_2 \otimes c_2$, $T_3 = u' \otimes (b_1 + b_2) \otimes c_3$: the pair (T_1, T_2) shares slot a ; solving in slot b gives $\lambda = 1$, and after the flip T_1 and T_3 share slot b exactly. Over $\mathbb{F}_2$ solved and random flips coincide in *value* ($\lambda = 1$) and differ only in *selection*; over $\mathbb{F}_p$ the solved λ is one specific field element sampling would never hit — the only reliable coincidence factory (measured, §1).

M-5. Closing move (second-alignment factory; ours). Given an *already aligned* pair, search for a single flip against any third summand that makes a second slot proportional, after which M-2 fires. Over $\mathbb{F}_p$ it needs coplanar triples in the transfer slot — generic states lack them (the structural reason naive-descent fails over big fields); over structured scheme neighborhoods it is routine (every storm quench).

M-6. Sandwich (symmetry) move (planned). Apply $(P, Q, R) \in \text{GL}(m, K)^3$: a gauge teleport. Flips are equivariant, so flip-graph components come in isomorphic families (K–M observe isomorphic components); the sandwich move merges each family into one search component. Cheap and strictly connectivity-increasing.

M-7. Closure jump (planned for $\mathbb{F}_p$; exists over $\mathbb{F}_2$ in **anf**). Fix two of the three factor tensors; the Brent system becomes m^2 independent linear systems in the third — solve exactly. When the current products are linearly independent the solution is unique (no move); when dependent, the solution space is positive-dimensional and the jump teleports across it. Toy: at rank 8 one split above Strassen, the eight products span a 7-dimensional space, so fixing α, β leaves a 1-parameter γ -family — the closure walks it in one step. The strongest field-generic non-local move we know (pure linear algebra).

M-8. Repair hop (built: `pursue8 --repair`). Delete k summands, beam-rebuild the residual in $\leq k - 1$ (record event) or k (lateral hop). Exhaustive over $\binom{r}{k}$ at small k — how the radius-rigidity certificates were bought (3×3: all 30 census seeds $k \leq 9$ rigid over two fields; 4×4: DPS-48 $k \leq 7$ rigid over Goldilocks, 73.6M subsets). As a walk move ($k = 2$ lateral) it hops between flip components at bounded cost.

M-9. Constructor move (AlphaTensor’s game; built: `pursue8`). Hold a residual tensor, subtract arbitrary rank-one terms until zero. Complete in principle — every scheme reachable — but unguided beam search stalls at the Strassen hump (3×3: terminal count 27; measured), precisely the gap a learned policy filled for DeepMind [4].

M-10. Completion jump (HKS method 2; built: `walk.py` over $\mathbb{F}_2$). Freeze a random subset of a known scheme’s bits, re-solve the rest with **stochastic local search (SLS)** — WalkSAT-family bit-flipping with noise over the Brent constraints (our native-ANF variant is detailed in the 53-schemes report). Non-local and powerful (our 53-scheme discovery; the ≈ 715 -scheme method-2 basin); field-restricted to where a solver exists — over $\mathbb{F}_p$ the analogue is M-7’s linear closure.

3 Search techniques tried (with verdicts)

technique	engine	verdict (measured)
random-walk storms	<code>flip23/48/23p</code>	81 new $\mathbb{Z}$ -classes; 63 field-novel patterns
exhaustive closure (BFS)	<code>--native/--lams</code>	exact component maps
depth-2 exhaustive fringe	<code>pursue5</code>	840 parents closed, 0 reductions
beam on nearmiss	<code>pursue6 chase</code>	8 levels, 1.45B states, $57\times$ null, 0 conversions
mix-and-quench	<code>pursue7</code>	~ 31 M walks over 3 fields; no records
constructor beam	<code>pursue8</code>	Strassen-hump stall at 27
exhaustive k -repair	<code>pursue8 --repair</code>	rigidity radii 9 / 7 / 6
persistent monotone walk	<code>pursue9</code>	$\mathbb{F}_2$: 64 $\rightarrow$ 53 then starvation; $\mathbb{F}_p$ sterile
frontier pool ladder	<code>pursue10</code>	$\mathbb{F}_2$: 64 $\rightarrow$ 49 in 3 min; running
SAT completion	<code>walk.py+anf</code>	53 schemes in 6 min; basin ≈ 715
native-ANF SLS	<code>anf</code>	8/10 HKS challenge-1
census-guided seeding	<code>SHORTLIST</code>	mobility metrics; band calibration
null-controlled method	<code>nmrand</code>	gradients real ($57\times$), not rank inflation

Planned next: M-6 sandwich moves, $\mathbb{F}_2$ solved-flip targeting, M-7 closure jumps over $\mathbb{F}_p$, M-8 as walk move, Moosbauer–Poole symmetric-subspace walks [2], commutative flipC, learned guidance (the neural track’s `gmi/gmi_mcgs` engines).

4 Classical search algorithms on this space

The rank hunt is single-agent search: **state** = gauge-canonical scheme; **actions** = the moves of §2; **goal** = any state of rank below the record (verification $O(m^6)$). The graph has heavy transpositions (hash canonical forms). Two cost models matter: *hop count* (every move 1) and *altitude* (splits 1, flips 0, reductions $-$).

BFS. Complete, optimal in hops, memory $O(\text{frontier})$. This *is* our `--native` closure and the fringe/flower enumerations — feasible when the reachable set is 10^6 -ish. Use for certificates, not hunting.

DFS / iterative deepening. $O(\text{depth})$ memory; our random walks are memoryless randomized DFS; principled depth-capped DFS appears in `cfloor`’s IDDFS (the adds bounds), where it is exact. For hunting, branching (10^2 – 10^4) dwarfs useful depth (10^5 +) : unattractive.

Uniform-cost / Dijkstra / SPF. Unit costs: it is BFS. Under the altitude model (splits 1, flips ε) UCS computes the *minimum-altitude descent* — “how high must any path climb to leave this basin” — the quantity the flower’s cut-vertex anatomy probes empirically. Frontier memory confines it to component scale, where it would upgrade “no crossing found at +8” to “no crossing exists below +9.”

A*. UCS plus admissible h . The obstruction is honest: an admissible h must lower-bound *path* distance to a smaller-rank scheme, and our rigorous bounds (flattenings, counting floors) bound the *tensor*, not the path. Practice therefore uses the inadmissible cousins: the chase is **greedy best-first** on nearmiss (h only), null-certified to carry signal ($57\times$) yet Goodhart-prone (zero conversions). A concrete admissible-within-radius candidate now exists: h built from the repair-ladder certificates (no scheme within replacement-distance k has smaller rank $\Rightarrow$ any descent needs $> k$ replacements).

Minimax and α/β . No adversary: minimax proper does not apply. But α/β ’s soul — cut a branch when a bound proves it cannot beat the incumbent — is **branch-and-bound**, and we already run it: the adds floors prune classes (“ ≥ 56 , stop”), the repair ladders prune subset-trees by projected completion size. The transferable idea: give the rank hunt the same treatment via certified component invariants.

MCTS / MCGS. Best fit, with precedent: AlphaTensor *is* MCTS with a learned prior on M-9’s space. Formulation: node = scheme (graph search with transposition table — MCGS); rollout = storm/pursue10 walk; reward = $-(\text{best rank reached})$; selection = UCT. Unguided MCTS must be compared at equal budget against best-of- K walks (our standing methodology note). The repository already carries MCGS scaffolding from the neural track (`gmi`, `gmi_mcgs`); the marginal build is the environment adapter, and the natural first target is the $\mathbb{F}_2$ 47 ladder, where ground truth exists and rollouts are cheap.

algorithm	fits?	as what	status
BFS	yes (bounded sets)	closures, certificates	in use
DFS/IDDFS	partially	cfloor sweeps; walks	in use
UCS/Dijkstra/SPF	yes (altitude cost)	min-altitude escape	possible
A*	blocked on admissible h	greedy best-first today	partial
weighted A*	yes	nearmiss + repair-radius h	planned
minimax	no adversary	—	n/a
α/β spirit (B&B)	yes	floors/repair pruning	in use
MCTS/MCGS	yes	guided constructor/ladder	scaffolding exists

5 One formulation to hold them all

Define the **layered scheme multigraph** $G(K, m)$: vertices are gauge-canonical schemes stratified by rank; intra-layer edges are flips (M-1, M-4, M-6, M-7); downward edges are reductions (M-2); upward edges are splits (M-3); non-local chords are repair and completion hops (M-8, M-10). Every technique of §3 is a traversal policy on G ; every algorithm of §4 is a standard policy under a cost model from {hops, altitude, compute}; every certificate we publish is an exactly-explored finite subgraph. The record hunt is: *find any vertex in a layer below the known minimum* — and §1’s disconnection results say the decisive question is not which traversal is cleverest but which *edge set* makes the target layer reachable. That is why the move inventory of §2, not the algorithm zoo of §4, is this note’s center of gravity.

Acknowledgments

This note was prepared in an extended interactive collaboration with Claude (Anthropic; Fable 5), which implemented the engines and drafted the text under the author’s direction and review.

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